

management of Sprague as he was their employee. That is the interest shown by Sprague for an invention, a method still used for IC manufacture. To quote Moore again, "Yeah, it's very much the same technology today."

In an article entitled 'Microelectronics,' published in 'Scientific American,' Robert Noyce wrote, "The integrated circuit, as we conceived and developed it at Fairchild Semiconductor in 1959, accomplishes the separation and interconnection of transistors and other circuit elements electrically rather than physically. The separation is accomplished by introducing p-n diodes or rectifiers, which allow current to flow in only one direction. The technique was patented by Kurt Lehovec at the Sprague Electric Company."

Noyce came up with a workable solution unaware that Jack Kilby at Texas Instruments had already succeeded, albeit with germanium. Silently working behind was Jean Hoerni, one of Fairchild's original founders, when he developed the 'planar' process. By using the planar process, each layer could now be isolated electrically. No need to cut the layers and join them where required as was done in the past. By mid 1959, Noyce created an IC made of silicon, using the cutting-edge insulating process developed by Jean Hoerni.

Gordon Moore confided in an interview, "In fact, when I look at the development of the integrated circuit, I always measure it from the first planar transistor rather than from the first integrated circuit."

Fairchild Semiconductor filed a patent for a semiconductor IC based on the planar process on July 30, 1959. But Texas Instruments had filed a comparable patent with Kilby's IC some time before. After a decade-long legal battle, the U.S. Court of Customs and Patent Appeals sustained Noyce's claims on interconnection techniques but gave Kilby and Texas Instruments credit for the first working IC.

Earlier, a German engineer Werner Jacobi (of Siemens AG) had filed a patent for an IC-like device. It was

a five-transistor amplifier designed to produce cheap hearing aids. Commercial use of his patent was not reported.

In a historical coincidence, Noyce and Kilby invented the IC without knowing each other and about the same time. Noyce's silicon IC is more efficient, more practical and the most common form now. NASA used Noyce's ICs for the first computers in the spacecraft of the Gemini programme.

By 1968, 'the most creative team' at Fairchild decided to start its own company. With initial capital from Arthur Rock, a venture capitalist, NM Electronics (NM standing for Noyce Moore) was incorporated on July 18, 1968 for developing large-scale ICs. Andrew Grove was roped in who would remain with them as president and CEO into the 1990s. The company's name was soon changed to Intel, taken from the first syllables of integrated electronics.

Just in a few months, Intel produced the 3101, a high-speed random access memory (RAM) chip. Those were the days when semiconductor memories were much more expensive than standard magnetic core memories. Intel felt that the future was in semiconductor memories which would soon replace magnetic cores.

Evolution of microprocessor

In a dramatic turn of events, in November 1971, Intel presented the 4004 to the public as "a new era of integrated electronics a micro-programmable computer on a chip." The dawn of the microprocessor! Gordon Moore called it, "one of the most revolutionary products in the history of mankind."

The invention of the microprocessor is a turning point in Intel's history and the industrial world. Interestingly, the development of 4004, the world's first microprocessor, was an offshoot of a necessity. Noyce quipped, "In a small town, when something breaks down, you don't wait around for a new part, because it's not coming. You

make it yourself." The glory is now with Ted Hoff.

Ted Hoff recalled in an interview, "We were contacted by a Japanese calculator company whose calculators came out under the name Busicomp. They said that they would like to have us build a family of chips for a whole series of different calculator models, models that would vary in type of display, whether they had a printer or not, the amount of memory that they had and so on.

While on the subject, let us digress and go back to Kilby once again. Patrick E. Haggerty, then TI chairman, challenged Kilby to design a calculator that could fit in a coat's pocket—equal or better than the bulky electro-mechanical desktop models available those days. Just to give you an idea, a calculator released just a year earlier weighed 55 pounds and cost \$2500. The result is the handheld calculator, of which Kilby is a co-inventor. He held about 60 patents including one for a thermal printer.

Now back to Ted Hoff; Busicomp contracted Intel to design cost-effective chips for a series of calculators. The project was assigned to Ted Hoff who did not like the idea which required 12 custom chips "because there was a lot of random logic and many interconnections between different chips."

In the words of Ted Hoff, "It seemed to me we could simplify the control logic, reduce the number of transistors and cut the overall cost.... Together Stan Mazor and I—Stan joined at the beginning of September—created an outline of what we were talking about and our marketing department proposed our alternate approach to the calculator company in the middle of September."

Hoff said, "Our initial goal was never to make a microprocessor, only to solve this particular customer's problem, this calculator design problem. But there were several aspects of the design that became more evident as it was pursued. One was, being more general-purpose and faster than the original design, we figured it might be